

Data Structures — Course Syllabus

Unit 1: Arrays and Linked Lists

An array is a collection of elements stored at contiguous memory locations, allowing constant-time access via indexing but requiring costly insertions and deletions. A linked list is a linear data structure where elements, called nodes, are linked using pointers; it allows efficient insertion and deletion but only sequential access. Variants include singly linked lists, doubly linked lists, and circular linked lists.

Unit 2: Stacks and Queues

A stack is a Last-In-First-Out (LIFO) data structure supporting push and pop operations, commonly used in function call management and expression evaluation. A queue is a First-In-First-Out (FIFO) data structure supporting enqueue and dequeue operations, used in scheduling and buffering tasks. Variants include circular queues, priority queues, and deques.

Unit 3: Trees

A tree is a hierarchical data structure consisting of nodes connected by edges, with one node designated as the root. A binary search tree (BST) maintains the property that left children are smaller and right children are larger than their parent, enabling efficient search, insertion, and deletion in $O(\log n)$ average time. Balanced trees like AVL trees and Red-Black trees maintain this efficiency in the worst case through rotation operations.

Unit 4: Sorting Algorithms

Quicksort is a divide-and-conquer sorting algorithm with average time complexity $O(n \log n)$ and worst-case $O(n^2)$, achieved by selecting a pivot and partitioning the array. Merge sort guarantees $O(n \log n)$ in all cases by recursively dividing the array and merging sorted halves. Other algorithms include bubble sort, insertion sort, and heap sort, each with different tradeoffs in time complexity and stability.